

Sumon Biswas

Assistant Professor at Case Western Reserve University
Department of Computer and Data Sciences, Case School of Engineering

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Computer science researcher focusing in the intersection of Software Engineering (SE) and Artificial Intelligence (AI). By adopting both formal and empirical approaches, I design verification and analysis of intelligent software systems for their fairness, safety, and sustainability.

Research Interest

Software Engineering	Design and Verification, Empirical Software Engineering, Coding AI Agents
Artificial Intelligence	Large Language Models, Algorithmic Fairness, Robustness
SE for AI	Safety of Agents, Causal Reasoning, Sustainable AI, Trustworthiness

Academic Qualifications

Postdoctoral Researcher <i>Carnegie Mellon University</i> Advisor: Dr. Eunsuk Kang	Pittsburgh, PA 2022–2024
Ph.D. in Computer Science <i>Iowa State University</i> Advisor: Dr. Hridayesh Rajan	Ames, IA 2016–2022
M.S. in Computer Science <i>Iowa State University</i>	Ames, IA 2016–2018
B.S. in Information Technology <i>Jahangirnagar University</i>	Dhaka, Bangladesh 2011–2015

Employments

Case Western Reserve University <i>Assistant Professor (tenure-track) at Department of Computer and Data Sciences</i> Case School of Engineering	Cleveland, OH Aug 2024 – p.
Carnegie Mellon University <i>Postdoc at School of Computer Science</i> Software and Societal Systems Department (S3D)	Pittsburgh, PA May 2022 – Aug 2024
Iowa State University <i>Research Assistant, Laboratory of Software Design</i>	Ames, IA May 2018 – May 2022
Iowa State University <i>Teaching Assistant, Department of Computer Science</i>	Ames, IA August 2016 – May 2018
Bangladesh University of Business and Technology <i>Lecturer, Department of Computer Science & Engineering</i>	Dhaka, Bangladesh January 2016 – July 2016

Publications

Referred Papers.....

- [1] Zhihao Dou, Qinqian Zhao, Zhongwei Wan, Dinggen Zhang, Weida Wang, Towsif Raiyan, Benteng Chen, Qingtao Pan, Yang Ouyang, Zhiqiang Gao, Shufei Zhang, [Sumon Biswas](#). *Plan Then Action: High-Level Planning Guidance Reinforcement Learning for LLM Reasoning*. International Conference on Machine Learning (ICML), 2026. [[Preprint](#)]
- [2] Sina Salimian, Gias Uddin, [Sumon Biswas](#), Henry Leung. *Bias Testing and Mitigation in Black Box LLMs using Metamorphic Relations*. In major revision, ACM Transactions on Software Engineering and Methodology (TOSEM), 2026. [[Preprint](#)]
- [3] Giang Nguyen, Shabbir Ahmed, [Sumon Biswas](#) and Hridesh Rajan. *FairSpec: Fairness Contracts for Machine Learning Pipelines*. In major revision, ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), 2026.
- [4] Zhihao Dou, Qinqian Zhao, Zhiqiang Gao, [Sumon Biswas](#). *ReShift: Aha-Moment-Driven Reasoning-Level Backdoor Attacks on Vision-Language Models*. Accepted in the European Conference on Computer Vision (ECCV), 2026.
- [5] Yining She, [Sumon Biswas](#), Christian Kästner and Eunsuk Kang. *FAIRSENSE: Long-Term Fairness Analysis of ML-Enabled System*. In Proceedings of the 47th IEEE/ACM International Conference on Software Engineering, ICSE 2025, page 782–794, 2025. [[DOI](#)] [[Preprint](#)] [[Artifact](#)]
- [6] David OBrien, [Sumon Biswas](#), Sayem Mohammad Imtiaz, Rabe Abdalkareem, Emad Shihab and Hridesh Rajan. *Are Prompt Engineering and TODO Comments Friends or Foes? An Evaluation on GitHub Copilot*, In Proceedings of the 46th IEEE/ACM International Conference on Software Engineering, ICSE 2024, page 1–13, 2024. [[DOI](#)]
- [7] [Sumon Biswas](#) and Hridesh Rajan. *Fairify: Fairness Verification of Neural Networks*. In Proceedings of the 45th IEEE/ACM International Conference on Software Engineering, ICSE 2023, page 1546–1558, 2023. [[DOI](#)] [[Artifact](#)]
- [8] Usman Gohar, [Sumon Biswas](#) and Hridesh Rajan. *Towards Understanding Fairness and its Composition in Ensemble Machine Learning*. In Proceedings of the 45th IEEE/ACM International Conference on Software Engineering, ICSE 2023, page 1533–1545, 2023. [[DOI](#)] [[Artifact](#)]
- [9] Giang Nguyen, [Sumon Biswas](#) and Hridesh Rajan. *Fix Fairness, Don't Ruin Accuracy: Performance Aware Fairness Repair using AutoML*. In Proceedings of the 31st ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering, ESEC/FSE 2023, page 502–514, 2023. [[DOI](#)] [[Artifact](#)]
- [10] [Sumon Biswas](#), Mohammad Wardat and Hridesh Rajan. *The Art and Practice of Data Science Pipelines: A Comprehensive Study of Data Science Pipelines In Theory, In-The-Small, and In-The-Large*. In Proceedings of The 44th International Conference on Software Engineering, ICSE 2022, page 2091–2103, 2022. [[DOI](#)] [[Artifact](#)]
- [11] David OBrien, [Sumon Biswas](#), Sayem Mohammad Imtiaz, Rabe Abdalkareem, Emad Shihab and Hridesh Rajan. *23 Shades of Technical Debt: An Empirical Study on Machine Learning Software*. In Proceedings of the 30th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering, ESEC/FSE 2022, page 734–746, 2022. [[DOI](#)] [[Artifact](#)]
- [12] [Sumon Biswas](#) and Hridesh Rajan. *Fair preprocessing: Towards understanding compositional fairness of data transformers in machine learning pipeline*. In Proceedings of the 29th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering, ESEC/FSE 2021, page 981–993, 2021. [[DOI](#)] [[Artifact](#)]
- [13] [Sumon Biswas](#) and Hridesh Rajan. *Do the machine learning models on a crowd sourced platform exhibit bias? an empirical study on model fairness*. In Proceedings of the 28th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering, ESEC/FSE 2020, page 642–653, 2020. [[DOI](#)] [[Artifact](#)]

Under Review.....

- [14] Jitong Zou and [Sumon Biswas](#). *Are VLMs Fair to Pedestrians? An Empirical Study on the Fairness of Vision-Language Models for Autonomous Driving*. Submitted, IEEE/ACM International Conference on Automated Software Engineering (ASE), 2026.
- [15] Alif Al Hasan and [Sumon Biswas](#). *What Breaks When LLMs Code? Characterizing Operational Safety Failures of Agentic Code Assistants*. Submitted, IEEE/ACM International Conference on Automated Software Engineering (ASE), 2026.

- [16] Zhihao Dou, Qinqian Zhao, Sumon Biswas. *AlgoForge: Specializing Code Generation Agents through Collaborative Reinforcement Learning*. Submitted, Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2026.

Referred Workshops.....

- [17] Sumon Biswas, Yining She and Eunsuk Kang. *Towards Safe ML-based Systems in Presence of Feedback Loops*. To appear in Proceedings of the 1st International Workshop on Dependability and Trustworthiness of Safety-Critical Systems with Machine Learned Components, **FSE-SE4SafeML** Position Paper, 2023.
- [18] Sumon Biswas, Md Johirul Islam, Yijia Huang, and Hridesh Rajan. *Boa meets Python: A Boa dataset of data science software in Python language*. In 2019 IEEE/ACM 16th International Conference on Mining Software Repositories, **MSR 2019**, 2019. [DOI]
- [19] Sumon Biswas, M. S. Kaiser, and S. A. Mamun. *Applying ant colony optimization in software testing to generate prioritized optimal path and test data*. In 2015 International Conference on Electrical Engineering and Information Communication Technology (ICEEICT), 2015. [DOI]
- [20] Manan Binth Taj Noor and Sumon Biswas. *A secure data security infrastructure for small organization in cloud computing*. In 2015 International Conference on Electrical Engineering and Information Communication Technology (ICEEICT), 2015. [DOI]
- [21] Sumon Biswas, Anisuzzaman, Tanjina Akhter, M. S. Kaiser, and S. A. Mamun. *Cloud based healthcare application architecture and electronic medical record mining: An integrated approach to improve healthcare system*. In 2014 17th International Conference on Computer and Information Technology (ICCIT), 2014. [DOI]

Grants and Proposals

Awarded Grants.....

CTSC Grant on AI-Assisted IPV Prevention

CWRU

Fairness Assessment and Improvement for AI-Enabled Systems for Dating Violence Prevention

2025

Awarded \$7,500 pilot grant to develop fairness-aware AI detection systems for dating violence (DV) prevention in youth digital communication, in collaboration with University Hospitals. The project curates representative datasets across youth subgroups, conducts algorithmic fairness audits of DV detection models, and prototypes bias mitigation strategies – serving as a foundational step toward a large-scale NIH proposal.

Grant Proposals under Review.....

NSF FutureCore

Case Western Reserve University

SHF: Characterizing and Repaying AI-Accelerated Technical Debt

2026

Collaborative NSF proposal (\$1M total; CWRU share: \$500K) to systematically characterize and address technical debt introduced or accelerated by AI-assisted software development. CWRU is the lead institute with Biswas as PI, and Tulane University is the collaborative institute with Dr. Hridesh Rajan as PI.

NIH R01 on AI-Based IPV Detection

Case Western Reserve University

AI-Enabled Health Risk Assessment for Intimate Partner Violence Survivors

2025

Collaborative NIH proposal for FAVORAP (Family Violence Health Risk Assessment Program), an AI-enabled clinical decision-support framework leveraging national and local EHR data to predict health risks for IPV survivors. Integrates co-morbidity network analysis, rule-based modeling, and graph-transformer methods, with emphasis on algorithmic fairness, cross-site generalizability, and stakeholder-guided pilot deployment.

Grant Proposal Contributions.....

DARPA V-SPELLS

Carnegie Mellon University | 2024

Verified Security and Performance Enhancement of Large Legacy Software 

Worked in the synergetic DARPA project to design compositional DSLs and check the safety and liveness of Unmanned Aerial Vehicles (UAV). In particular, building model extraction technique from legacy source code and then verify the models for safe drone systems, e.g., Ardupilot, MAVSDK.

NSF EAGER

Carnegie Mellon University | 2023

Towards a Design Methodology for Software-Driven Sustainability 

Through this grant, we proposed a framework for sustainability as a *first-class software quality attribute*. I contributed to writing the proposal and currently building a simulation platform to generate sustainable design strategies for ML-enabled systems.

NSF TRIPODS

Iowa State University | 2019 – 2022

D4: Dependable Data-Driven Discovery 

I actively contributed writing this grant proposal and thereafter it received the institute-level grant under HDR TRIPODS by NSF. I assisted my advisor (who is the PI of the grant), to launch the interdisciplinary data science hub at Iowa State under this institute.

NSF CNS – Boa

Iowa State University | 2021

Enhancing Infrastructure for Studying Software and its Evolution at a Large Scale 

By enabling program analysis on machine learning programs and Jupyter notebooks, I contributed to this grant. Built the Python language support for Boa for large-scale software repository mining and analysis.

Facebook Probability and Programming Award

Iowa State University | 2020 – 2021

Big code assisted neural architecture search 

Contributed to writing the grant proposal and thereafter my dissertation was partially supported by this industry grant. My work on compositional reasoning of ML pipeline (FSE'21) was funded by the grant.

Research Mentoring

Present

Ph.D.	Alif Al Hasan ◦ Jitong Zou ◦ Zhihao Dou Case Western Reserve University
M.S.	Panimalar Gobichettipalayam Annadurai ◦ Towsif Raiyan ◦ Ruksaar Shaik Case Western Reserve University
Undergraduate	Khue Luong ◦ Ohta Kamiya ◦ Anika Kaur ◦ Sheila Monera Cabarique ◦ Ali Nawaf ◦ Maximillian Schulten Case Western Reserve University
High School	Sharon Sharma K-12 Intern, Case Western Reserve University

Past

2025	Zelan Eroz Espanto Case Western Reserve University (Undergraduate student)
2023 – 2025	Sayan Nath (he/him) University of Calgary, M.S. (thesis) student
2023 – 2025	Sina Salimian (he/him) University of Calgary, M.S. (thesis) student
2022 – 2024	Yining She (he/him) Carnegie Mellon University (Ph.D. student)
2021 – 2023	Giang Nguyen (he/him) Iowa State University (Ph.D. student)
2020 – 2022	David OBrien (he/him) University of Northern Iowa (B.S.) → Iowa State University (Ph.D. student)
2020 – 2022	Usman Goher (he/him) Iowa State University (Ph.D. student)
2019 – 2020	Senior Design Project Team Department of Electrical and Computer Engineering, Iowa State University

Honors & Awards

- **Distinguished Reviewer:** Selected as *Distinguished Reviewers* for the journal, ACM Transactions on Software Engineering and Methodology (TOSEM)  2022
- **Research Excellence Award:** Awarded by Iowa State University recognizing outstanding research 2022

- **ACM CAPS Award:** Awarded by ACM SIGSOFT to attend in-person ICSE 2023 conference held at Pittsburgh, PA 2022
- **Publication Award:** Awarded by Computer Science Department at Iowa State University for publication in the top-tier venues in consecutive years 2021
- **Panelist at ESEC/FSE'20:** Selected as a panelist in the session on Fairness at ESEC/FSE 2020 [↗](#) 2020
- **PLMW Scholar in PLDI'19:** Awarded by Programming Language Mentoring Workshop (PLMW) at the PLDI conference at Phoenix, AZ 2019
- **Professional Advancement Grants:** Awarded by Graduate and Professional Student Senate at Iowa State University 2019
- **ACM Travel Grant:** Awarded to attend IMA Workshop organized by Institute for Mathematics & its Applications (IMA), University of Minnesota [↗](#) 2019

Services

Federal Agency Service

- **NSF AI Institute Site Visitor:** Served on the NSF Site Visit Team Panel at the University at Buffalo for the NSF AI Institute on AI and Speech Language Therapy. 2026
- **NSF Panelist:** Served as a panelist for the National Science Foundation (NSF) proposal review panel for the Division of Computing and Communications Foundations (CCF) 2025

Organizing Committee

- **LLMTrust Workshop:** General Chair of the International Workshop on Trustworthy Large Language Models for Software Engineering (LLMTrust) [↗](#) 2026
- **WiDS Worldwide Conference Event:** Co-organized the daylong Women-in-Data-Science (WiDS) event at Iowa State University as part of the annual WiDS Worldwide Conference 2022
- **SPLASH'21 (Hybrid):** Served as the Accessibility Chair of the ACM SIGPLAN conference on Systems, Programming, Languages, and Applications (SPLASH 2021) with OOPSLA and REBASE [↗](#) 2021
- **Web Chair:** Served as the web chair of the Midwest Big Data Summer School, organized by Iowa State University, May 17-20, 2021 [↗](#) 2021
- **SPLASH'20 (Virtual):** Served as the Accessibility Chair of the ACM SIGPLAN conference on Systems, Programming, Languages, and Applications (SPLASH 2020) with OOPSLA, ECOOP and REBASE [↗](#) 2020

Journal Reviewer

Publons Web of Science reviewer profile [↗](#)

- **TOSEM:** Serving in the *Board of Distinguished Reviewer* for ACM Transactions on Software Engineering and Methodology (TOSEM) [↗](#) 2021-p
- **TSE:** Serving as the reviewer for IEEE Transactions on Software Engineering (TSE) 2021-p
- **EMSE:** Serving as the reviewer for Springer Journal - Empirical Software Engineering (EMSE) 2022-p
- **IEEE Software:** Serving as the reviewer for IEEE Software. 2022-p
- **Affective Computing:** Serving as the reviewer for IEEE Transactions on Affective Computing 2024-p
- **InfSof:** Serving as the reviewer for Information and Software Technology (InfSof) 2024

Conference Program Committee (PC)

- **FSE'26 - Technical Track:** Serving in the program committee of FSE 2026 Technical Track [↗](#) 2026
- **ICSE'25 - Technical Track:** Serving in the program committee of ICSE 2025 Technical Track [↗](#) 2025
- **ASE'25 - NIER:** Serving in the program committee of ASE 2025 NIER Track [↗](#) 2025
- **ICSE'24 - Technical Track:** Served in the program committee of ICSE 2024 Technical Track [↗](#) 2024
- **ASE'24 - Technical Track:** Served in the program committee of ASE 2024 Technical Track [↗](#) 2024
- **ASE'23 - Technical Track:** Served in the program committee of ASE 2023 Technical Track [↗](#) 2023
- **ASE'23 - Industry Showcase:** Served in the Industry Showcase track of ASE 2023, Luxembourg [↗](#) 2023
- **ASE'22 - Industry Showcase:** Served in the Industry Showcase track of ASE 2022, Oakland, MI [↗](#) 2022
- **ASE'22 - LBR:** Served in the Late Breaking Results (LBR) track of ASE 2022, Oakland, MI [↗](#) 2022

- **ESEC/FSE'22 - DS**: Served in the Program Committee of Doctoral Symposium track of ESEC/FSE 2022, Singapore [↗](#) 2022
- **MSR'21 - Technical Track**: Shadow PC member of the Technical Track of International Conference on Mining Software Repositories (MSR'21) held at Madrid, Spain [↗](#) 2021
- **OOPSLA'21 - AEC**: PC Member of Artifact Evaluation Committee (AEC) at ACM SIGPLAN conference on Object-Oriented Programming, Systems, Languages & Applications (OOPSLA) held at Chicago, IL [↗](#) 2021

Teaching

Instructor

CSDS 393/493 - Software Engineering Class size: ~80
Undergraduate/Graduate level, Case Western Reserve University Spring 2025, Spring 2026

Topics SE lifecycle models; team organization and project management • Requirements analysis and specification • Software design techniques and architecture • Implementation and programming practices • Software testing, validation, and analysis • Software maintenance; SE for and with AI; ethics and open source

CSDS 447 – Responsible AI Engineering Class size: ~32
Graduate level, Case Western Reserve University Fall 2025

Topics Responsible AI requirements and assurance (fairness, robustness, safety, explainability) • Bias detection and mitigation; adversarial evaluation; hallucination handling • Practical trade-offs in responsible AI; hands-on team project on real-world AI-enabled systems

CSDS 600 - Special topics: Responsible AI Engineering Fall 2024
Graduate level, Case Western Reserve University

Topics Fairness, robustness, safety and interpretability

Lecturer

Guest Lecturer Class size: ~40
Graduate level, Iowa State University October 2023

Course Responsible AI: Risk Management in Data-Driven Discovery

Lecture title Assessing and Ensuring Fairness in ML Models

Topics covered Understanding societal bias and AI fairness • Fairness metrics and mitigation • Fairness verification • Sustaining fairness

Primary Instructor Class size: ~50
Bangladesh University of Business and Technology January 2016 - July 2016

Responsibilities Lecture • Design courses • Create tests • Conduct programming lab sessions

Courses **1. Computer and Programming Concepts**: Instruct two sections: computer science major and non-computer science major

2. Structured Programming Language: Instruct theory and lab sessions

3. Pattern Recognition: Instructor for the seniors in computer science major

Talks

1. **Kent State University**: Gave an invited talk “Engineering Responsible AI: From Fairness to Long-term Impact” at the Robotics and Autonomous Systems (AUTOBOT) Program at the College of Aeronautics and Engineering, Kent State University, 2025 Kent, OH
2. **Amazon**: Presented my research on “Fairness verification and debugging” at the Automated Reasoning Group in AWS, 2024 New York
3. **Oracle Labs**: Presented my research on “Mining and analysis of AI-Enabled Software Systems” in the Oracle Labs, 2024 Boston
4. **SE4SafeML Workshop**: Presented the position paper “Towards Safe ML-based Systems in Presence of Feedback Loops” in the workshop of ESEC/FSE 2023 San Francisco

5. **DARPA PI Meeting:** Presented “*Software Architecture for Unmanned Aerial Vehicles (UAV): Automatic AADL Model Extraction and Reasoning*” in the DARPA V-SPeLLS PI Meeting, 2023 Virtual
6. **ICSE’23:** Presented the technical track paper entitled “*Fairify: Fairness Verification of Neural Networks*” at the 45th ACM/IEEE ICSE conference, 2023 Melbourne
7. **ICSE’23:** Presented the technical track paper entitled “*Towards Understanding Fairness and its Composition in Ensemble ML*” at the 45th ACM/IEEE ICSE conference, 2023 Melbourne
8. **Invited Talk:** Invited by Dr. Wallapak Tavanapong (Professor and Chair of Broadening Participation, Iowa State University) to present my research on “*Fairness Verification and Sustainability of AI Systems*”, 2023 Virtual
9. **ICSE’22:** Presented technical track paper entitled “*The Art and Practice of Data Science Pipeline*” at the 44th ACM/IEEE ICSE conference, 2022 Pittsburgh, PA
10. **Invited Talk:** Presented my research in the CREATE SE4AI group at NSERC, Canada; participated by students and researchers from Concordia University, Polytechnique Montreal, Queen’s University, and University of Alberta. Talk title: “*Understanding and Reasoning Fairness of ML-Based Software*” (February 2022) [↗](#) Virtual
11. **NSF PI Meeting:** Presented my work on “*Modular DNN Verification*” in the NSF TRIPODS PI meeting chaired by Dr. Lenore J. Cowen from Tufts University, December 2021 Virtual
12. **ESEC/FSE’21:** Presented technical track paper entitled “*Fair Preprocessing*” at the 29th ACM ESEC/FSE, 2021 Athens, Greece
13. **ESEC/FSE’20:** Presented technical track paper entitled “*An empirical study on ML Model Fairness*” at the 28th ACM ESEC/FSE, 2020 Sacramento, CA
14. **TADS Presentation:** Presented my research on “*Fairness of Machine Learning Models*” in front of Theoretical and Applied Data Science (TADS) Group of D4 Institute, Iowa State University, 2020 [↗](#) Ames, IA
15. **MSR’19 Presentation:** Data showcase and paper presentation entitled “*Boa meets python: A boa dataset of data science software in Python language*” in Mining Software Repository Conference (MSR), 2019 [↗](#) Montreal, Canada

Research Highlights.....

- 2025 **Small Research Goes Big: When Less Represented Topics Find Resonance across an Entire Field:** Presented my research journey for the paper–“*The Art and Practice of Data Science Pipelines*” on the podcast series of the world’s largest network for academic audio. [↗](#)
- 2020 **Dependability:** My association and contributions towards the NSF TRIPODS grant have been highlighted by the department. [↗](#)

Activities

Professional Development.....

- **Learning Fellows Program:** Accepted as a UCITE Learning Fellow at Case Western Reserve University. 2025
- **New Faculty Symposium:** Attended the New Faculty Symposium at ICSE 2025 in Ottawa, Canada. 2025

Outreach.....

- **Research Outreach:** Mentored Zihe Zhao (she/her), an undergraduate student from Rice University (later MS student at CMU), in the ICSE 2023 Student Mentoring Workshop (SMW) on graduate study and research career trajectory. <https://conf.researchr.org/track/icse-2023/icse-2023-student-mentoring-workshop> 2023
- **SIGPLAN Blog:** Co-authored the ACM SIGPLAN blog to share the experience of organizing a premier conference in hybrid mode for the first time. <https://blog.sigplan.org/2022/08/25/hybrid-splash-2021-retrospective> 2022
- **Ambassador of Women-in-Data-Science:** Served as the Ambassador of Women in Data Science (WiDS) initiative organized by Stanford University. [↗](#) 2022
- **Lunch-n-Learn:** Organized the weekly interdisciplinary research discussion for the year at Iowa State Theoretical and Applied Data Science (TADS) group. 2020-21
- **K-12 Outreach:** Organized and served as the judge of the K-12 computational project competition at Iowa State University. 2018

Affiliation

- **ACM:** Professional Member of Association for Computing Machinery (ACM) 2019-*p*
- **SIGSOFT & SIGPLAN:** Member of the ACM SIGSOFT and ACM SIGPLAN 2020-23
- **IEEE Computer Society:** Member of the IEEE Computer Society 2023